

Final ISP NOC Platform Database Schema

Stack: Flutter + FastAPI (Python) + PostgreSQL

Core Tables

users

id, username, password_hash, full_name, email, created_at

locations

id, location_name, location_type, latitude, longitude, parent_location_id, created_at

devices

id, device_name, hostname, vendor, device_type, ip_address, location_id, parent_device_id, status, last_seen, created_at

interfaces

id, device_id, interface_name, interface_type, status, rx_traffic, tx_traffic, speed, last_updated

device_status

id, device_id, status, latency_ms, packet_loss, checked_at

device_metrics

id, device_id, cpu_usage, memory_usage, temperature, ping_latency, packet_loss, collected_at

wireless_metrics

id, device_id, signal_strength, noise_floor, ccq, tx_rate, rx_rate, collected_at

pppoe_sessions

id, customer_name, username, router_id, session_ip, mac_address, uptime, status, connected_at, disconnected_at

alerts

id, device_id, alert_type, severity, message, status, created_at, resolved_at

outages

id, location_id, root_device_id, outage_type, affected_customers, severity, status, started_at, resolved_at

notifications

id, alert_id, notification_type, recipient, message, status, sent_at

event_logs

id, device_id, event_type, message, severity, created_at

olts

id, device_id, vendor, management_ip, location_id, status

pon_ports

id, olt_id, port_number, status

onus

id, pon_port_id, serial_number, customer_name,
optical_power, status

towers

id, device_id, location_id, tower_name, status

sectors

id, tower_id, sector_name, frequency,
capacity, status

Final ER Relationships

USERS

LOCATIONS

■■■■ DEVICES

■ ■■■■ INTERFACES

■ ■■■■ DEVICE_STATUS

■ ■■■■ DEVICE_METRICS

■ ■■■■ WIRELESS_METRICS

■ ■■■■ ALERTS

■ ■■■■ EVENT_LOGS

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■■■■ OLTS

■ ■■■■ PON_PORTS

■ ■■■■ ONUS

■

■■■■ TOWERS

■■■■ SECTORS

DEVICES

■■■■ PPPOE_SESSIONS

ALERTS

■■■■ NOTIFICATIONS

OUTAGES

■■■■ ROOT_DEVICE

MVP Build Order

1. users
2. locations
3. devices
4. interfaces
5. device_status
6. device_metrics
7. pppoe_sessions
8. alerts
9. outages
10. notifications
11. event_logs
12. topology
13. OLT/ONU
14. Towers/Sectors